

8.11 LINE SIZING CRITERIA

Line sizing is performed for rated capacity. Unless there are specific Technology Licensor requirements, the line sizing shall be carried out as per criteria provided in the following sections.

8.11.1 Line Sizing Criteria for Liquids

SERVICE	LINE SIZE	MAXIMUM VELOCITY m/s	PRESSURE DROP (Kg/cm ²)/km	
			Normal	Max.
Pump suction, bubble point (1)	≤2"	0.6	0.6	0.9
	3" – 10"	0.9		
	12" – 18"	1.2		
	≥ 20"	1.5		
Pump suction, subcooled	≤2"	0.9	2.3	3.5
	3" – 6"	1.2		
	8" – 18"	1.5		
	≥ 20"	1.8		
Pump discharge, P ≤ 50 Kg/cm ² (g)		1.5 to 3 (8)	3.5	4.5
P > 50 Kg/cm ² (g)			7.0	9.0
Gravity flow		0.6 (3)	0.25	0.45
Side-stream draw-off (2)	≤ 2"	0.6	0.6	0.9
	≥ 3"	0.9		
Cooling water (7): i. sub-header		2.5	2.5	3.5
ii. main header			0.6 – 1.5	(4)
iii. Between pumping station and unit		1.5 to 3	1.5	(4)
Kerosene jet fuel		(5) 3.0 (max)		
Hot oil		1.0 (min)		
Lean Amine (velocity lamination max.)		2.0 for (CS) / 5.0 for SS		
Rich Amine (velocity lamination max.)		1.0 for (CS) / 5.0 for SS		
Caustic (velocity lamination max.)		1.5 for (CS)		

Notes:

- (1) Applicable to liquid containing dissolved gas.
- (2) Provide a vertical run of 3 meters minimum from nozzle, at nozzle size, before reducing the size of the line.
- (3) Normal velocity.
- (4) To be analyzed case by case.
- (5) 50 to 100 m upstream tank inlet or loading facilities, the velocity shall be reduced to 1 m/s limit risks associated with static electricity.
- (6) Pressure Drops for liquid to thermosiphon reboiler are 0.2 – 0.4 (Kg/cm²)/km to be checked by heat exchanger section.
- (7) Minimum velocity is 1 ~ 1.2 m/sec.
- (8) For off sites large lines velocity criteria can be increased

8.11.2 Line Sizing Criteria for Gas & Vapors

	ρv^2 (max) Pa	Pressure Drop (Kg/cm ²)/km	Maximum Velocity m/s
1. VACUUM SERVICE		4 % Abs. Press. Max.	90
2. COMPR. SUCT.		0.2 – 0.7	
3. COMPR. DISCH. GAS		0.4 – 1.0	
4. STEAM (sub-headers)	15 000		
a) 1 Kg/cm ² (g)		0.4 – 1.0	(1)
b) 10 – 40 Kg/cm ² (g)		1.0 – 2.0	
c) > 40 Kg/cm ² (g)		(3)	
5. STEAM (long lines)	15 000		(1)
a) 1 Kg/cm ² (g)		0.1 – 0.2	
b) , c) > 10 Kg/cm ² (g)		0.2 – 1.0	
6. KETTLE REBOILER OR NATURAL CIRCULATION RETURN LINE		0.2 – 0.4 (2)	
7. OVHD VAPOUR FROM STRIPPER		0.2 – 0.45 (2)	
8. COLUMN OVHD (P ≥ atm.)	15 000	0.3 – 0.6	
9. COLUMN OVHD (vacuum)		TOTAL = 5 mm Hg max	90
10. GAS LINES			(2)
P ≤ 20 Kg/cm ² (g)	6 000		
20 < P ≤ Kg/cm ² (g)	7 500		
50 < P ≤ 80 Kg/cm ² (g)	10 000		
P > 80 Kg/cm ² (g)	15 000		

Notes:

(1) The following table applies to set the velocity:

Pipe size	Saturated Steam	Superheated Steam
< 2"	10 m/s	15 m/s
3" – 8"	30 m/s	40 m/s
> 10"	40 m/s	60 m/s

(2) For large diameter lines of over 300 mm, it should be verified that the following velocities are not exceeded

Pressure (Kg/cm ² (g))	Max. Velocity (m/s)
0.1 or lower	60.0
Up to 1.0	40.0
Up to 2.5	30.0
Up to 10.0	15.0
Up to 20.0	10.0
Up to 40.0	7.0

(3) To be analyzed case by case

8.11.3 Erosional Services

	Pressure Drop (Kg/cm ²)/km	Velocity m/s
1. Mixed Phase Condensates	0.2 – 0.3	10 – 20
2. Reboiler Return Line (Natural Circulation)	0.2 – 0.4	
3. Partial Condenser Outlet	0.3 – 0.6	
4. Mixed Phases at Compressor Delivery	0.4 – 1.0	

Vertical and horizontal pipes should not be affected by slug flow.

In a first step, the following criteria can be used

$$v_m = 10 \text{ to } 23 \text{ m/s}$$

$$\rho_m v_m^2 = 5\,000 \text{ to } 10\,000 \text{ Pa (15\,000 Pa Max)}$$

where:

$$\rho_m = \text{density of mixed phase}$$

$$v_m = \text{velocity of mixed phase (equal to 10 to 23 m/s)}$$

8.11.4 High Velocity Services

In offshore facilities piping handling gaseous and mixed phase (gas / vapor + liquid) streams, velocity shall be lower than the erosional velocity defined in API RP-14E/3, that is:

$$V_e = \frac{C}{\sqrt{S}}$$

Where:

V_e = ft/s

S = lbs/ft³

C = refer to the following table

“C” VALUES	
FOR CONTINUOUS SERVICE	
150	For high alloy/corrosion resistant material.
100	For uniform corrosion with rates lower than 0.3 mm/a, or when a corrosion inhibitor is used.
80	For uniform corrosion with rates higher than 0.3 mm/a.
FOR DISCONTINUOUS SERVICE	
200	For all cases.

Velocity shall be in any case lower than 50 m/s in continuous services except vacuum services, while for discontinuous services (as safety valve discharges to flare) velocity shall be lower than:

- MACH for individual users 0.75
- MACH for flare header 0.50

In case of solid particle entrainment, limitations for discontinuous services shall be used.

8.11.5 Liquid & Gas with Solids

	Velocity (m/s)
1. Liquid + solids	$1 \leq v \leq 3$ (1)
2. Gas + solids (pneumatic conveying)	$20 \leq v \leq 30$ (1)

Notes:

1. For licensed plants, refer to Licensor requirements.